

Anugayathiri Pugazhenthir

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Authorized to work in the U.S. without sponsorship

Summary

- Experienced Embedded Linux System Software Engineer with hands-on expertise in kernel compilation, U-Boot, root filesystem customization, and integration of third-party libraries tailored to system requirements.
 - Skilled in developing and automating functional test cases in C, and proficient with C/C++ and object-oriented programming (OOP) concepts.
 - Extensive experience with UNIX/Linux interprocess communication mechanisms, including shared memory, pipes, and signals for embedded systems
 - Developed multithreaded applications and worked on protocol design and implementation for Bluetooth file transfer.
 - Familiar with source control tools such as SVN and Git/GitHub for version management and collaboration.
 - Proficient in analyzing business processes and translating them into technical designs to drive embedded application development.
 - Implemented Linux hardening and security features for embedded devices to ensure system robustness.
 - Experienced in cross-compilation for ARM architectures and device driver development, including printer and I/O drivers.
 - Worked on software architecture, design, and testing of real-time, multi-threaded embedded applications using C and Linux technologies.
 - Strong knowledge of embedded real-time operating systems (RTOS), synchronization, and memory management.
 - Effective collaborator with cross-functional teams, including hardware engineers, QA, and system architects, to deliver production-ready solutions.
 - Passionate about entrepreneurship and leveraging embedded technology expertise to build innovative products and startups.
 - Excellent communication, teamwork, and problem-solving skills with a keen eagerness to learn and apply new technologies.
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Skills

- Languages & Tools: C, C++, Python, Linux, Git, Bash scripting
 - Embedded Systems: Embedded Linux, Bootloader, Kernel Development, Root File System, OTA Updates, RTOS Fundamentals, Device Driver Interfaces, Board Bring-Up Experience, OS Internals Knowledge, Hypervisor Knowledge, ARM Cores Experience, Debugging Tools, and Electronic Test Equipment.
 - Debugging & Analysis: GDB, JTAG (Trace32), UART console, perf, kernel tracing
 - Build & Toolchain: Cross-compilation, GCC, CMake, Git, Jenkins
 - Hardware: FPGA (Altera DE2-115), CAN, SPI, I2C, TI (Texas Instruments) platforms
 - Machine Learning: Naive Bayes, Logistic Regression, SVM (Applied in Hardware Security Research)
 - Software: Altera Quartus II, ModelSim
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Education

Master of Science in Embedded Electrical and Computer Systems

San Francisco State University — December 2020

Bachelor of Engineering in Electronics and Communication

Anna University — September 2011 – May 2015

Professional Experience

Embedded Linux System Software Engineer

Lucid Motors | March 2022 – September 2024

- Developed and maintained **embedded firmware and low-level software** for an ARM-based automotive platform (TI TDA4VM), supporting ADAS camera systems.
- Worked on **bootloader and system bring-up**, including boot flow from ROM → U-Boot → Linux, ensuring reliable system initialization and hardware readiness.
- Implemented and debugged **device driver-level interactions** and hardware interfaces, including UART, I2C, and SPI, for peripheral communication and sensor integration.
- Wrote efficient, production-quality code in C/C++ for performance-critical embedded components, focusing on memory safety and real-time behavior.
- Performed **low-level debugging and system validation** using UART logs, JTAG, and hardware debugging tools to diagnose boot failures and runtime issues.
- Contributed to **OTA update system** with A/B partitioning, including image validation and fail-safe mechanisms for reliable firmware upgrades.
- Automated firmware flashing, logging, and validation using **Python and shell scripting**, improving bring-up efficiency and reducing manual effort.
- Collaborated cross-functionally with hardware and perception teams to support **sensor integration and system-level debugging**.
- Worked with **ARM Cortex-A and MCU subsystems**, with an understanding of multi-core interactions and system-level resource management.

Design Verification Engineer

CloudBig Technology | March 2021 – February 2022

- Gained hands-on experience in design and verification engineering, aligned with academic background in Embedded Systems.
- Created verification environments and participated in pre-silicon functional verification at block, chip, and system levels.
- Contributed to CPU and graphics core verification using SystemVerilog and UVM methodologies.
- Designed and validated synthesizable RTL code, and supported performance modeling and post-silicon validation.
- Strengthened skills in real-time debugging, verification tools, and design engineering best practices.

Software Engineering Intern

Kinetic Reality | October 2020 – December 2020

- Developed and tested motion detection algorithms for real-time embedded systems.
- Collaborated with the core engineering team to design and optimize motion-sensing software.
- Applied principles of signal processing to enhance motion tracking accuracy.

Research Assistant – Hardware Security Lab

San Francisco State University | June 2018 – December 2020

- Investigated the use of DRAM-PUF startup values for unique ID generation and assessed their security vulnerabilities.
- Applied machine learning algorithms to analyze ID resiliency.
- Presented findings at the 2018 International Green and Sustainable Computing Conference.
- Published a paper in the 2019 SPIE Defense and Commercial Sensing Conference.

[Research Link](#)

Teaching Assistant – Digital System Design Lab (ENGR-378)

San Francisco State University | January 2019 – June 2019

- Assisted students in hands-on labs with Altera DE2-115 FPGA boards and VHDL-based design.
- Supported the use of **Quartus II** and **ModelSim** for simulation and synthesis in Digital System Design coursework.

Bluetooth Voice-Controlled Bot

Academic Project | <https://adp-bot.webnode.com/>

EEG Eye-Blink Controlled Bot

Academic Project | <https://nmi-project-ads.webnode.com/>